

Reasoning RL Primer

Deep Dive into DeepSeek — Interlude

Yin Yang

Foundation Books Project

The one rule of this interlude

Training can reinforce only what the reward *exposes*.

- A pretrained policy receives a prompt, samples several completions, and receives scores for properties a *declared* grader can observe.
- Training then makes higher-scoring samples more likely.
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Goal: establish that one loop — sample, grade, compare, update, limit the change — and the vocabulary around it, before the V3 post-training and R1 chapters add recipe, equations, and implementation.

1. Why reinforcement learning here

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2. One loop before the terminology

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3. Three layers that must stay separate

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4. Translations that keep their boundaries

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5. Misconceptions and the handoff

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So reinforcement learning can *sample* alternatives, *score* the visible outcomes, and move probability toward the better samples.

Note the asymmetry that makes this work: the grader has to recognize a good answer, not produce one.

What automatic scoring does not remove

Warning (Task design does not disappear)

Automatic scoring replaces some target-writing work with prompts, extractors, verifiers, sampling rules, reward coverage, and movement guards.

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Every number in this interlude's worked example is a teaching fixture. [OPEN: private prompt mixture and reward weights]

One loop before the terminology

The congruence fixture, priced by a local grader

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The mean reward is 0.55: a comparison baseline, not a passing score. [DERIVED: this primer's teaching fixture]

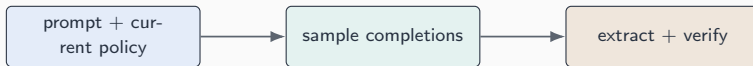
One iteration of the loop

prompt + cur-
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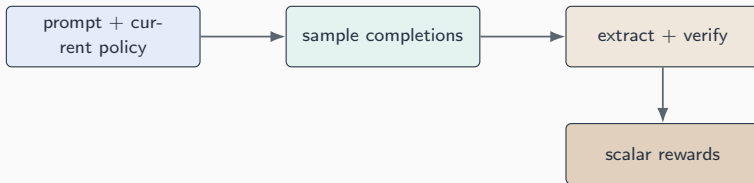
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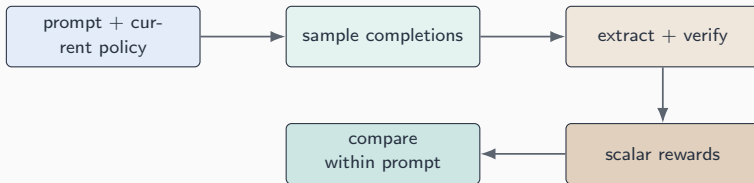
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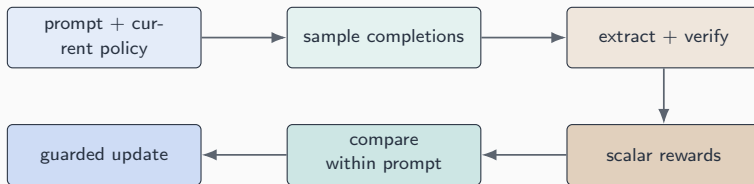
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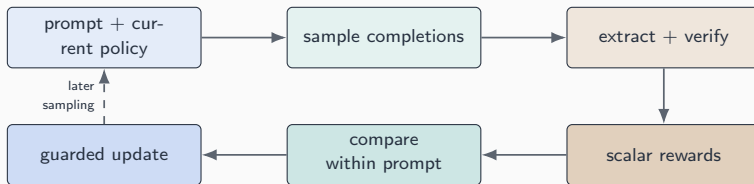
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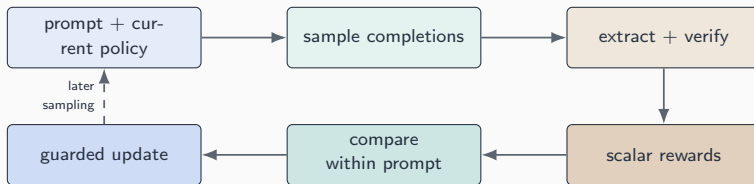
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The grader sees *declared output properties*, not every property of the reasoning. The update changes future sampling probabilities; it does not paste a response into the model.

If every sampled reward is equal, the comparison stage has no relative direction to pass to the update — whether the answers were all right or all wrong.

**Three layers that must stay
separate**

Reward, comparison, movement control

Layer	What it does	What it cannot do
Task and reward	Extractor and verifier assign credit for the properties they test	Calling a reward an objective does not complete its specification

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Kullback–Leibler divergence measures distribution drift on a declared probability surface. It does not check mathematical correctness.

The PPO bridge: clipping credits, it does not constrain

PPO separates two questions: *which* sampled completions receive update pressure, and *how much objective credit* the current-to-old policy ratio receives.

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Actor–critic PPO commonly learns advantages with a critic and Generalized Advantage Estimation. GRPO replaces that *source* with within-prompt comparison while keeping a PPO-shaped clipped-ratio guard. [REPORTED: DeepSeekMath, arXiv:2402.03300]

**Translations that keep their
boundaries**

Classical RL, bounded translation

Classical term	Language-model reading	Main-path use
State or observation	Prompt plus generated prefix	Conditions the next-token distribution

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Tool-using or multi-turn agents can add several interactions and rewards. The reasoning chapters begin with terminal completion rewards because that is what their examples require.

Three post-training signals, three evidence paths

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For the congruence prompt: SFT would imitate one worked answer, RL would compare the four sampled completions, and distillation would learn from a stronger teacher's response or distribution.

Pipelines can combine all three. Combining them does not make them interchangeable, and does not erase their distinct evidence paths.

Misconceptions and the handoff

Shortcuts to drop before the equations

Shortcut	Correction
Reward proves truth. Advantage is answer quality. The reference policy supplies the answer. Old and reference policies have one job.	Reward certifies only the grader's tested properties. Advantage compares reward with a baseline. It anchors drift; the reward supplies task feedback. The old policy anchors a ratio; the reference policy anchors a drift penalty.
Pure RL starts from random weights.	R1-Zero starts from a pretrained base; <i>pure</i> marks no preliminary cold-start SFT in that account.
A rollout must be physical.	Here, a sampled text completion is a rollout.
RL creates missing capability on demand.	Results depend on the starting policy, prompts, exploration, and reward coverage.

From concept to code: what “pure” actually marks

The release itself states the starting point and the ordering, which is what makes the “random weights” shortcut checkable rather than arguable.

from DeepSeek-R1/README.md

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“Without SFT as a preliminary step” is a statement about *ordering*, not about initialization: the base model is pretrained in both accounts. R1 then adds the cold-start stage that R1-Zero omits. [REPORTED: DeepSeek-R1 README.md]

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This primer stops at conceptual vocabulary and qualitative update direction; it does not specify an executable optimizer. Chapter 7 separates SFT, preference evidence, RL, and distillation. Chapter 8 begins the mathematical and implementation track. Chapter 9 adds the cold-start, refinement, distillation, and release-facing stages.